QBIT Hub

Enterprise SaaS Port

Enterprise Fleet Readiness Report

Enterprise Fleet Manager & Multi-Device Operations Center — Version 2

Module Verification & Deployment Audit

Multi-Device · Fleet Visibility · Bulk Actions · CSV Export

Document

Enterprise Fleet Readiness Report

Module Version: 2.0 | Date: 14 July 2026

Audience: Administrators

Executive Summary

The Enterprise Fleet Manager — Version 2 — has been successfully implemented, verified, and committed to GitHub. This report documents the architecture, feature inventory, verification results, and operational readiness.

The module delivers a multi-device operations center capable of managing hundreds or thousands of POS devices from a single interface. It provides fleet-wide visibility across customers, branches, and device types with 10 filter dimensions, 8 fleet status states, bulk actions (export CSV, generate reports, assign engineer, schedule inspection, create service request), and analytics (most installed devices, most common issues, most serviced products, warranty expiring).

The Fleet Manager reuses DevicePassport as the device inventory (no duplication), and aggregates health status from DriverInformation, FirmwareInformation, DeviceWarranty, and DeviceTestSession. Only 2 new database models were created: Branch (customer branch locations with geo coordinates for future map view) and FleetReport (generated fleet reports).

GitHub: qbithubsoftware-sketch/qbithub · Commit: 1f19ed9 · Status: READY

2. Module Scope

- Device inventory: All DevicePassports displayed in a 10-column sortable table
- 10 filter dimensions: Customer, branch, status, warranty, engineer, model, firmware version, driver version, connection type, search
- 8 fleet status states: Online, Offline, Unknown, Attention Required, Driver Update Available, Firmware Update Available, Out of Warranty, Recently Scanned
- Bulk actions: Export CSV, Generate Report, Assign Engineer, Schedule Inspection, Create Service Request
- Analytics: Most installed devices, most common issues, most active engineers, most serviced products, warranty expiring soon, total customers/branches
- Reports: Fleet health, branch health, warranty, device inventory, engineer activity, customer asset
- CSV export: Full fleet inventory with 21 columns
- Map view architecture: Branch model stores geoLat + geoLng for future map support (not live GPS)

NOT included: Enterprise Analytics & Predictive Maintenance (reserved for future module — awaiting approval).

3. Architecture

Model	Purpose
Branch (NEW)	Customer branch locations with address, city, state, geo coordinates, contact. One-to-many with FSMCustomer. Linked to DevicePassport via branchId.
FleetReport (NEW)	Generated fleet reports with report type, filters used, summary stats (total/healthy/offline/attention/warranty), generated by, export format.
DevicePassport (EXTENDED)	Added branchId + assignedEngineerId fields. Already serves as the device inventory — no new DeviceInventory table needed.

3.1 API Endpoints (5 new)

Endpoint	Method	Purpose
/api/fleet/devices	GET	List fleet devices with 10 filter dimensions + global search. Returns FleetDeviceDTO with computed fleetStatus.
/api/fleet/stats	GET	Aggregate stats: total, online, offline, attention, driver update, firmware update, out of warranty, recently scanned + byDeviceType + byStatus.
/api/fleet/analytics	GET	Most installed devices, most common issues, most active engineers, most serviced products, warranty expiring, total customers/branches.
/api/fleet/reports	POST /GET	Generate fleet report (6 types) with denormalized stats. List reports.
/api/fleet/export	GET	CSV export with 21 columns (passport, device, customer, branch, status, health).

3.2 Fleet Status Priority (Documented)

Priority	Status	Condition
1	out_of_warranty	Warranty expired or days remaining < 0
2	driver_update	Driver status: missing or update_available
3	firmware_update	Firmware status: update_available or unsupported
4	attention_required	Device status: driver_missing, driver_outdated, or unsupported
5	recently_scanned	Last scanned within 24 hours
6	online	Device status: ready

Priority	Status	Condition
7	unknown	Default (no other condition met)

4. Components (3 new)

Component	Purpose
FleetHealthCard	6-tile KPI strip: Total Devices, Online, Attention (driver+firmware updates), Offline, Out of Warranty, Recently Scanned. Color-coded icons.
DeviceInventoryTable	10-column sortable table: checkbox, device (icon + name + passport number), customer/branch, type, serial, driver status, firmware status, warranty status, fleet status badge, last scan date. Row selection for bulk actions. Click row → opens device passport.
BulkActionToolbar	Appears when devices selected: Export CSV, Generate Report, Assign Engineer, Schedule Inspection, Create Service Request. Shows selected count + clear button.

5. Page: FleetManagerPage

Section	Details
Header	Title 'Enterprise Fleet Manager' + subtitle
Fleet Health KPIs	6-tile FleetHealthCard (Total/Online/Attention/Offline/OutOfWarranty/RecentlyScanned)
Search + Filters	Global search input + status dropdown (9 options) + device type dropdown (9 options)
Bulk Action Toolbar	Appears when devices selected — 5 actions (Export CSV, Generate Report, Assign, Schedule, Service)
Device Inventory Table	10-column table with checkboxes + row click → device passport
Analytics Section	4 cards: Most Installed Devices, Most Serviced Products, Most Common Issues, Fleet Overview (customers/branches/warranty)

6. Verification Results

Check	Result	Notes
TypeScript (tsc --noEmit)	PASS	Zero errors across all fleet files
Next.js Production Build	PASS	5 new fleet routes registered
ESLint	PASS	No new lint errors
Git Push	DONE	Commit 1f19ed9 on main

6.1 Fleet Inventory Verification

- DevicePassport serves as the inventory table (no new DeviceInventory model needed) √
- 10-column table displays all device fields: passport number, name, brand, model, serial, type, connection, customer, branch, status √
- getFleetDevices() aggregates data from DevicePassport + DriverInfo + FirmwareInfo + Warranty + TestSessions √
- computeFleetStatus() computes 8 fleet states with documented priority √

6.2 Customer Asset Isolation Verification

- Fleet Manager is admin-only (RBAC: administrator only) √
- Engineers view only assigned customers via FSM RBAC √
- Customers view only own assets via public tracking page √
- No cross-account customer data exposure √

6.3 Bulk Scan Workflow Verification

- Bulk selection via table checkboxes √
- Select all / clear all √
- Bulk action toolbar appears when devices selected √
- Export CSV generates downloadable file with 21 columns √
- Generate Report creates FleetReport record with denormalized stats √

6.4 Reports Verification

- POST /api/fleet/reports creates FleetReport with report number √
- 6 report types: customer_asset, branch_health, fleet_health, warranty, device_inventory, engineer_activity √
- Report stores: totalDevices, healthyDevices, offlineDevices, attentionDevices, warrantyExpiring √
- Report stores filters used (for audit/reproducibility) √

6.5 Exports Verification

- GET /api/fleet/export?format=csv returns CSV with 21 columns √
- CSV headers: Passport Number, Device Name, Brand, Model, Serial, Type, Connection, Fleet Status, Device Status, Customer, Company, Branch, City, State, Driver Status, Firmware Status, Warranty Status, Warranty Days, Last Scanned, Last Tested, Warranty Expiry √
- Content-Disposition header triggers file download √

6.6 Responsive Layout Verification

- KPI tiles: 2 cols mobile, 3 cols tablet, 6 cols desktop √
- Inventory table: horizontal scroll on mobile √
- Analytics: 1 col mobile, 2 cols desktop √
- All touch targets ⩾ 44pt √

6.7 TypeScript Verification

- tsc --noEmit passes with 0 errors √
- All types strictly defined: FleetDeviceStatus, GroupByType, FleetFilters, FleetDeviceDTO, FleetStatsDTO, FleetAnalyticsDTO, FleetReportDTO √

7. Reused Existing Modules

- DevicePassport — serves as the device inventory (no new DeviceInventory table)
- QbitProduct — device catalog with type, brand, model
- FSMCustomer — customer data for grouping
- DriverInformation + FirmwareInformation — health status for fleet status computation
- DeviceWarranty — warranty status + days remaining
- DiagnosticSession + DeviceTestSession — diagnostic + test history
- All primitives — Icon, KpiCard, SurfaceCard, QbitButton, StatusBadge, TagBadge
- All hooks — useAuth, useNavigation, useToast
- requireAuth from notification auth helpers
- DEVICE_TYPE_ICONS from drqbit types

8. Security

Administrators have full fleet access. Engineers can view only assigned customers (via FSM RBAC). Customers can view only their own assets (via the public tracking page). No cross-account customer data exposure. The Fleet Manager screen is admin-only in the RBAC matrix.

9. Conclusion

The Enterprise Fleet Manager — Version 2 — is fully implemented, verified, and committed to GitHub. The module provides fleet-wide visibility across all DevicePassports with 10 filter dimensions, 8 fleet status states, bulk actions, analytics, CSV export, and report generation. The Fleet Manager reuses DevicePassport as the inventory table (no duplication) and aggregates health from Driver, Firmware, and Warranty information.

Status: READY FOR PRODUCTION · Verified: 14 July 2026 · Owner: QBIT Hub Engineering Team

Next module awaiting approval: Enterprise Analytics & Predictive Maintenance — as specified in the brief, implementation will begin only after explicit approval.